

Muhammad Farooqi

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EDUCATION

University of Waterloo

Mechatronics and Robotics Engineering + AI Option — Grade: **91%**

Relevant Courses: OOP in C/C++, Data Structures and Algorithms, Digital Logic, Linear Circuits, Linear Algebra.

Waterloo, ON

Expected Apr-2028

TECHNICAL SKILLS

Programming Languages: Python, C, C++, TypeScript, JavaScript, SQL, HTML/CSS

AI & ML: PyTorch, OpenCV, NumPy, TensorRT, YOLOv8/v11, HuggingFace, ChromaDB, MediaPipe

Robotics & Simulation: ROS2, Isaac Sim, Foxglove, Rviz, Gazebo, Isaac Lab, URDF, SLAM, Arduino, Raspberry Pi

Developer Tools: Git/GitHub, Docker, Ubuntu/Linux, Jupyter Notebook, React.js, Node.js, Figma, Ollama

EXPERIENCE

Navigation Software Engineering Intern

Sep 2025 – Present

Rivian

Vancouver, BC

- Owned integration test planning for the vehicle Driver Display swap feature, creating **40** test cases covering ADAS and map view transitions, button press sequences, and map state changes, catching **5** critical blockers before release.
- Pioneered early merge-request checks by triggering **downstream CI** jobs in a separate test repository, preventing navigation-breaking changes that previously occurred in **10–20%** of daily builds.
- Designed vehicle nav **API tests** for reroutes and trip refresh across R1T, R1S and R2 vehicles, covering **10** scenarios.
- Implemented **20** Appium based end-to-end navigation tests for reroutes, destination edits, and driver display changes, running nightly in **CI pipelines** and catching **15** navigation regressions during new feature development.

Perception Software Engineer

Feb 2025 – Present

WATOnomous

Waterloo, ON

- Built a **3D voxel** perception pipeline using **PyTorch** and **NVIDIA spconv** library and projective geometry, converting RGB-D images to point clouds and sparse voxel grids for robotic perception.
- Developed a **ROS2** perception node in Python for RGB-D fusion and sparse **3D** voxelization, achieving **4×** runtime speedup via stride-based downsampling and depth-range filtering
- Dockerized ROS2** 3D vision stack to enable reproducible **GPU-accelerated** training and inference workflows.

Robotics Software Engineer Intern

May 2025 - Aug 2025

Lincoln Electric Automation

Waterloo, ON

- Designed and deployed an end-to-end tack welding program for 6-axis **Fanuc** (KAREL) and **ABB** (RAPID) arms, removing separate welder setup and saving **5–7 seconds** per tack, **2–3 minutes** per part across **2** production cells.
- Developed motion control and weld parameter interfaces in **TypeScript** using **functional programming**, integrating with **Controller APIs** for 3 tack sizes and validating through **100+** virtual welds in **RobotStudio** and **RoboGuide**.
- Configured **RenovateBot** in **Azure DevOps** to auto-update dependencies, reducing outdated packages by **80%**.

Backend Software Developer Intern

Sep 2024 – Dec 2024

ISARA

Waterloo, ON

- Built cryptographic risk analysis algorithms for TLS, SMB, and plaintext protocols using **Node.js** and **GraphiQL**.
- Collaborated with QA to resolve **355+** failures during regression testing, improving product health by **88.75%**.
- Automated demo deployment using a **Python** script to fetch build from an **AWS S3** bucket and developed an admin control panel with **React**, **PostgreSQL**, and **Docker**, saving **10 minutes** per deployment.

Computer Vision Intern

Jan 2024 – Apr 2024

AirMatrix

Mississauga, ON

- Annotated over **3,000 traffic images** using **Roboflow**, achieving **95%** accuracy at **0.5 IoU** on vehicle detection.
- Trained a **YOLOv8 vehicle detection model** in **Python** for vehicle detection, counting, and speed estimation.

PROJECTS

SLAM ROS2 Robot | *Linux, ROS2, Python, Gazebo, Rviz, Camera/Lidar, Raspberry Pi, Arduino*

- Created **URDF** files with collision and inertia values for realistic simulation in **ROS2** and **Gazebo**.
- Configured a differential drive robot in **Rviz** with **camera** and **2D LiDAR** nodes, generating point clouds from LiDAR and applying **multi-sensor** fusion with **SLAM Toolbox** for mapping and localization.
- Integrated motor control via **Raspberry Pi 4** and **Arduino** over **UART** at **57600 baud**, and enabled teleoperation with a **PS4 controller** via the **joy** topic using the **ros2 control** library.

